GridSight EV: Geospatial EV Charging Infrastructure Intelligence Platform

IIT Bombay Techfest 2026–27 | Round 1 Submission Document | Challenge 1: EV Infra Map India

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1. Problem Statement, Solution Framework & System Overview

India's accelerated transition to electric mobility has outpaced public infrastructure planning tools, leaving investments vulnerable to severe capital misallocation, queuing bottlenecks, and localized grid overloads. Public charging planning currently faces four structural barriers:

- **Multi-Registry Fragmentation & Unit Ambiguity:** Official charging portals report conflicting metrics across disjointed data schemas. The Bureau of Energy Efficiency (BEE) national portal catalogs individual charger/connector rows (39,641 rows; 44,066 connectors across 29,371 sites); the Ministry of Petroleum and Natural Gas (MoPNG) tracks Oil Marketing Company (OMC) retail fuel outlets (27,737 installed; 22,753 operational); the Ministry of Heavy Industries (MHI) logs scheme-subsidized FAME-II deployments (8,932 OMC units); while OpenStreetMap (OSM) captures a sparse crowdsourced subset (725 national features; 179 in Delhi). Blindly aggregating these registries produces catastrophic double-counting.
- **Spatial Granularity Mismatch:** Operational EV registrations from MoRTH/VAHAN are aggregated at national, state, and Regional Transport Office (RTO) administrative levels, whereas charging and battery swapping stations operate at discrete point coordinates. In Delhi, commercial EV fleets (e-rickshaws, commercial cabs, delivery freight) register predominantly at the centralized Burari authority (DL-11/12/13/14), severely distorting localized demand.
- **Opacity of Critical Electrical Telemetry:** Public registries completely omit distribution transformer (DT) spare capacity, feeder voltage ratings (11 kV vs 33 kV), and real-time session utilization logs.
- **Geographic Proximity vs. Volumetric Adequacy Disconnect:** Conventional EV dashboards rely on naive circular coverage buffers ($d \leq r$), conflating spatial presence with capacity adequacy. A charging plaza offering two slow 3.3 kW AC sockets within 1.0 km cannot service 100 commercial e-rickshaws requiring 6.4 kWh/day each, causing queue collapse. GridSight EV explicitly decouples spatial accessibility from daily energy throughput (kWh/day).

GridSight EV resolves this planning challenge through an end-to-end geospatial intelligence architecture structured into four linked pillars: (1) Deterministic Multi-Source Integration with an uncompromising Non-Additivity Rule (registries are evaluated as independent, toggleable baselines); (2) High-Resolution Spatial Gap Engine covering NCT Delhi via a 375-cell planning grid (2.0 km × 2.0 km) integrating WorldPop demography, 3,534 OSM commercial activity attractors, and 8,087 road network segments; (3) Explainable Physics-Informed Spatial AI Suite comprising a Supervised Random Forest Demand Regressor ($R^2 = 0.796$), Bass Diffusion adoption forecasting ($R^2 = 0.992$), and a Multi-Objective Feeder-Constrained Siting Optimizer; and (4) Bottom-Up Capacity Planning Engine (2026–2030) modeling vehicle-class energy balances, public reliance, battery swapping, and physical hardware sizing (Stations, EVSE Points, AC/DC Nozzles, Swapping Dockets).

Platform Tier	Core Processing Engine	Methodological Function & Technical Deliverables
Tier 1: Data Ingestion & Sanitization	scripts/process_data.py (Python, ~1,770 lines)	Deterministic state normalization (46 raw strings → 36 ISO codes), coordinate sanitation (119 bad rows isolated), non-additivity layer isolation.
Tier 2: Spatial Harmonization Engine	scripts/process_data.py + WorldPop / OSM Extracts	Delhi 375-cell grid (2km × 2km), WorldPop 2020 scaled to NCP 2025 projection (22.3M), Burari commercial fleet activity redistribution crosswalk.
Tier 3: Explainable AI & Optimization	scripts/train_ai_models.py (Scikit-Learn, SciPy)	Random Forest regressor (120 trees, $R^2 = 0.796$), Bass Diffusion S-curve adoption model ($R^2 = 0.992$), Submodular greedy feeder-constrained siting optimizer.
Tier 4: UI & Capacity Simulator	src/utils/planningEngine.ts + React 18 / TypeScript / Leaflet	Sub-millisecond client-side spatial simulation, Sigmoid decay kernel, 2026–2030 fleet capacity sizing, zero-watermark keyless OSM cartography.

IMPLEMENTATION STATUS: OPERATIONAL BASELINE VS. PLANNED ROADMAP

• Implemented & Tested in Code/Data [Phase 0 - Verified Operational]: 36 States/UTs national demand choropleths (10.69M active EVs across 2 VAHAN fuel tags); BEE national registry geocoding (29,371 sites); Delhi 375-cell spatial grid (22.3M pop); Random Forest regressor; Bass Diffusion S-curve; Submodular greedy siting optimizer (top 25 plazas); Automated QA suite (28/28 assertions pass in scripts/validate_processed_data.py); Zero-watermark Leaflet UI.

• Planned Roadmap [Post-Round 1]: Highway corridor buffers (Phase 1, NH-48 / Yamuna Expwy); CEA secondary distribution transformer voltage tier proxy (Phase 2); DISCOM nodal interconnect live API ingest & containerized deployment (Phase 3).

2. Data Integration, Normalization & Spatial Methodology

Multi-Source Normalization & De-duplication: Non-Additivity Design Rule: Because BEE, MoPNG OMC (PIB), MHI FAME-II, Switch Delhi, and OSM overlap extensively but lack shared foreign keys, GridSight EV strictly maintains them as independent analytical baselines. The user interface allows planners to toggle one source baseline at a time, forbidding naive arithmetic aggregation that corrupts public policy planning.

Dual VAHAN Fuel Tag Reconciliation: Querying only the ELECTRIC(BOV) fuel tag in VAHAN captures 6.35M vehicles but omits 4.34M vehicles cataloged under PURE EV. GridSight EV deterministically integrates both tags across 36 States/UTs and 23 Delhi RTOs (10,689,490 active EVs nationwide; 549,153 in Delhi). Furthermore, our pipeline extracts active compliant (7.87M) versus active non-compliant registrations (2.82M, 26.4% lacking valid fitness/PUC) to establish upper- and lower-bound operational demand bounds.

Burari Commercial Fleet Spatial Redistribution: Commercial EV fleets (3W e-rickshaws/autos, 4W commercial cabs, buses, light goods) registered at Delhi's centralized Burari authority (DL-11/12/13/14) do not operate solely in North Delhi. GridSight EV redistributes these vehicles across Delhi's 9 Census districts using an empirical activity-weighted crosswalk, while private two-wheelers remain assigned to their zonal residential RTOs:

[Eq. 1] $w_{district} = 0.5 \times (N_{poi,district} / N_{poi,total}) + 0.5 \times (P_{district} / P_{total})$ — where N_{poi} is OSM commercial attractors (3,534 hubs) and P is Census district population (22.3M).

Spatial Grid Definition & Boundary Scope: Delhi 375-Cell Spatial Planning Grid: The spatial analysis grid spans bounding box 28.40°N–28.89°N, 76.83°E–77.35°E with uniform steps of 0.018° Lat × 0.0205° Lon (~2.0 km × 2.0 km at 28.6°N), generating 375 discrete cells. The bounding box explicitly captures the NCT plus commuter fringes (Gurugram, Noida, Faridabad, Ghaziabad) where cross-border fleet traffic concentrates. Demography is derived from WorldPop 1km raster scaled to the National Commission on Population (NCP) 2025 projection of 22,300,000 residents.

Coverage Assessment Logic: Two-Stage Coverage & Volumetric Adequacy Engine: Stage 1 assesses Euclidean/equirectangular distance d to the nearest operational charging station (distortion <0.05% vs Haversine across Delhi, <2 m/km). Cells are classified into: (a) Covered: $d \le r$ (user radius: 0.5 to 5.0 km; default 1.5 km); (b) Underserved Priority Zone: $d > r$ AND AI Demand Score $D \ge 22$ (65th percentile threshold); (c) Low-Demand Fringe: $d > r$ AND $D < 22$. Stage 2 evaluates volumetric adequacy: even if a cell satisfies $d \le r$, if installed daily throughput (kWh/day) falls below localized daily fleet demand, the cell flags an Energy Capacity Deficit.

DELHI SPATIAL PILOT SPECIFICATIONS & CARTOGRAPHIC ACCURACY BOUNDS

- High-Resolution Multi-Modal GIS Layers: 3,534 commercial attractors (fuel, transit, parking, market clusters) and 8,087 major road segments (motorway, trunk, primary, secondary) ingested via Overpass API and spatially indexed to the 375 grid cells.
- Equirectangular Planar Distortion Bound: Latitude scaling cos(28.6°) = 0.878; verified maximum planar distance error is <0.05% (<2 m per 1,000 m) compared to Great-Circle Haversine, enabling sub-millisecond client-side spatial queries without GIS engine lag.
- NCT Boundary + Gateway Buffer: Incorporates high-density logistics and commuter transfer nodes across border toll plazas (Kapashera, Badarpur, Kundli) where cross-border commercial fleets enter the capital network.

3. Explainable Spatial AI, Forecasting & Capacity Sizing Engine

1. Supervised Machine Learning Regressor: Supervised Spatial Demand ML Regressor: Rather than adopting arbitrary heuristic scoring weights, GridSight EV deploys a Random Forest Regressor (120 estimators, maximum tree depth 6) trained on the 375 planning cells against observed charging station density from 876 verified Switch Delhi stations. The model achieves $R^2 = 0.796$ (5-fold cross-validation $R^2 = 0.333 \pm 0.096$, MAE = 5.85). The lower cross-validation R^2 transparently reflects real-world supply-demand spatial decoupling—historical station siting followed municipal land availability rather than pure demand density. Feature importances derived from the trained regressor establish objective spatial weights: Population Density (49.65%), Commercial Activity POIs (13.14%), Commercial Attractor Hubs (9.53%), Distance to Feeder Substation (9.14%), Major Road Density (8.32%), Public Transit Interchanges (6.69%), and Local Substation / DT Count (3.55%).

[Eq. 2] $D_AI = f_RF(Pop, POI_comm, Transit, Roads, Dist_substation, DT_count) \in [0, 100]$ — Trained on 375 cells against 876 ground-truth stations (MAE = 5.85).

2. Continuous Distance-Decay Formulation: Continuous Sigmoid Distance-Decay Priority Kernel: Conventional planners use discontinuous step-functions where priority jumps abruptly from 0 to 0.65 D at distance r. GridSight EV implements a continuous 4th-order Sigmoid distance-decay formulation:

[Eq. 3] $S = D_AI \times [\, 1 - 1 / (1 + (d / r)^4) \,]$ — Smooth decay: $S \to 0$ when $d \ll r$; $S = 0.50\ D_AI$ at $d = r$; $S \to D_AI$ when $d \gg r$.

3. Bass Diffusion Fleet Growth Forecasting: Predictive Fleet Forecasting via Bass Diffusion Model: To model non-linear market adoption through 2030, GridSight EV calibrates the Bass Diffusion Model on 68 months of historical MoRTH VAHAN registrations (2018–2026) using Non-linear Least Squares ($R^2 = 0.992$):

[Eq. 5A] $S(t) = dN(t)/dt = [\, p + (q / M)\, N(t) \,] [\, M - N(t) \,]$ — Calibrated: p = 0.0114 (innovation), q = 0.4963 (imitation), M = 20.0M national ceiling.

The calibrated Bass model projects: (a) 2026 Baseline: 11.23M national EVs (P10: 8.16M, P90: 14.54M); Delhi active fleet: 577,295 EVs. (b) 2028 Mid-Term: 15.63M national EVs (P10: 12.23M, P90: 18.71M); Delhi active fleet: 803,300 EVs. (c) 2030 Target: 18.17M national EVs (P10: 15.27M, P90: 20.58M); Delhi active fleet: 933,751 EVs.

4. Daily Energy Balance & Infrastructure Sizing: Bottom-Up Daily Public Energy Demand Balance: Total public charging energy demand across vehicle classes $k \in \{2W, 3W, 4W, Bus, Goods\}$ is calculated as:

[Eq. 5B] $E_k = V_k,2030 \times e_k \times \varphi_k,s$ **(kWh/day)** — where $V_k,2030$ is projected fleet, e_k is daily consumption, and φ_k,s is public charging reliance.

Vehicle Class	Daily km	Specific Energy	Daily kWh (e_k)	Public Reliance (φ)	Charging / Swap Split
2-Wheelers (2W)	32 km/day	0.032 kWh/km	1.024 kWh/day	20% (80% home)	35% Swap / 55% AC / 10% DC
3-Wheelers (3W)	85 km/day	0.075 kWh/km	6.375 kWh/day	55% (45% depot)	35% Swap / 55% AC / 10% DC
4W Commercial/Cab	48 km/day	0.145 kWh/km	6.960 kWh/day	32% (68% private)	0% Swap / 35% AC / 65% DC Fast
e-Buses	180 km/day	1.150 kWh/km	207.00 kWh/day	20% (80% depot)	0% Swap / 0% AC / 100% DC Fast
Light Goods (LCV)	90 km/day	0.150 kWh/km	13.500 kWh/day	50% (50% depot)	0% Swap / 25% AC / 75% DC Fast

Hardware Sizing & Power Substitution Matrix: Daily throughput ratings are established as: AC Connector = 4.2 kW × 5.5 h = 23.1 kWh/day; DC Fast Nozzle = 30.0 kW × 6.0 h = 180.0 kWh/day; Battery Swap Docket = 12.5 swaps/day × 2.1 kWh = 26.25 kWh/day. Required EVSE Points = AC + ⌈ DC / 1.35⌉ ; Stations = ⌈ Points / 4.5⌉ [Eq. 6]. To prevent low-power sockets from superficially masking fast DC deficits, GridSight EV enforces a Power Substitution Matrix: Net Hardware = max(0, Required - ∑ Installed × η_power) [Eq. 7], where η_power = 1.0 if installed power matches or exceeds target tier, η_power = 0.15 for slow AC against slow AC targets, and η_power = 0.00 for slow AC against DC fast targets.

4. Empirical Findings, Siting Optimization & Automated Validation

Empirical Supply-Demand Discoveries: National & Delhi Baseline Findings: Across 36 States/UTs, active EV penetration stands at 3.84% (10,689,490 active EVs out of 278.4M registered vehicles). Four states account for 41.3% of all active EVs: Uttar Pradesh (2,364,570; 9.94% penetration, driven by rural/semi-urban e-rickshaws), Maharashtra (1,064,280; 3.02%), Delhi (549,153; 6.64%), and Karnataka (498,390; 3.65%). In Delhi, 876 verified operational Switch Delhi sites provide 2,437 charging points and 303 swapping stations. Under default r = 1.5 km, geographic coverage appears high (68.3% of urban cells within 1.5 km of a charger), yet volumetric adequacy reveals severe local deficits: commercial 3W and freight delivery clusters in Central, Shahdara, and outer North-West Delhi suffer up to 14,200 kWh/day unserved peak deficits.

Constrained Siting Optimizer: Multi-Objective Grid-Constrained Siting Optimizer: GridSight EV incorporates a greedy submodular optimization algorithm solving for the Top K optimal candidate infill charging plazas. The objective maximizes cumulative unserved AI demand score: max ∑_{i ∈ selected} S_i, subject to: (1) DISCOM Feeder Distance Constraint: Candidate cells with substation distance > 1.5 km are penalized; cells are stratified into electrical tiers: Tier 1 Feeder Ready (<1.0 km distance, ≥500 kVA spare proxy), Tier 2 Moderate (1.0–2.5 km, 150–250 kVA proxy), and Tier 3 Constrained (>2.5 km). (2) Spatial Anti-Cannibalization Constraint: Selected candidate hubs must maintain a minimum dispersion distance ≥ 1.2 km from one another. Local infill hardware guarantees site redundancy: minimum 2 AC sockets and 1 DC fast nozzle per selected hub (Eq. 4B).

Rank	Candidate Location / Ward	Grid Coordinates	Unserved Score (S)	Substation Dist	Electrical Feeder Tier
1	Burari Transit Transport Hub	28.751°N, 77.199°E	88.4 / 100	410 m	Tier 1 (Feeder Ready, ≥500 kVA)
2	Bawana Industrial Corridor	28.796°N, 77.035°E	82.1 / 100	620 m	Tier 1 (Feeder Ready, ≥500 kVA)
3	Okhla Phase-III Logistics Hub	28.535°N, 77.271°E	79.6 / 100	510 m	Tier 1 (Feeder Ready, ≥500 kVA)
4	Anand Vihar ISBT / Kaushambi	28.647°N, 77.312°E	76.3 / 100	780 m	Tier 1 (Feeder Ready, ≥500 kVA)
5	Najafgarh Freight Exchange	28.611°N, 76.984°E	71.9 / 100	1,120 m	Tier 2 (Moderate, 150–250 kVA)

AUTOMATED VERIFICATION & QUALITY AUDIT (28/28 TESTS PASSED)

- Data Integrity & AI Pipeline (scripts/validate_processed_data.py): 28 of 28 automated assertions PASSED (100% success). Verified: 36 states/UTs, 10.69M national active EVs, 39,641 BEE rows (44,066 connectors), 22,753 operational OMC stations, 8,932 FAME-II chargers, 549,153 Delhi EVs across 23 RTOs, 876 Switch Delhi sites, 375 grid cells (scaled to 22.3M pop), Random Forest R^2 = 0.796, Bass R^2 = 0.992.
- TypeScript Strict Static Typing: tsc --noEmit exited cleanly with 0 compilation errors across all frontend and engine components.
- Production Build & Live Smoke Test: Vite production build succeeded in 7.68s; local HTTP 200 OK test verified zero console errors and zero tile watermarks.

5. Data Strategy, Provenance & Methodological Caveats

All data sources were frozen as of 19 October 2026, stored in raw format under data/raw/, and tracked in data/source_manifest.csv:

ID	Source Agency	Vintage	Granularity	Scope in Platform	Known Limitations & Mitigation
SRC-01	BEE, MoP	Oct 2025	Point / Site	39,641 rows, 44,066 connectors across 29,371 sites	Static PDF; lacks operational status; 119 corrupted coordinates isolated.
SRC-02	MoPNG / OMC	Mar 2026	State / PSU	22,753 operational OMC retail outlet stations	OMC outlets only; kept separate from BEE to avoid double-counting.
SRC-03	MHI / FAME-II	Mar 2026	State / Scheme	8,932 operational FAME-II OMC charging units	Scheme-specific; isolated from general registries.
SRC-04	Switch Delhi	Oct 2026	Point / Site	876 deduplicated stations (573 charge + 303 swap)	15 duplicate kiosks purged; 121 swap kiosks geocoded via PIN centroid.
SRC-05	OpenStreetMap	Oct 2026	Point / Node	725 national stations; 179 in Delhi	Severe crowdsource undercount (2.5% of BEE). Used strictly as cross-check.
SRC-06	MoRTH / VAHAN	Oct 2026	State / RTO	10,689,490 active EVs (dual fuel tags: BOV + Pure EV)	Includes 26.4% non-compliant fleet; Burari distortion redistributed.
SRC-07	DataMeet Maps	2019 / 2011	Polygon	36 States/UTs and 9 Delhi Census districts	Official spatial administrative boundaries for GIS joins (CC BY 4.0).
SRC-08	WorldPop / NCP	2020 (2025)	1km Raster	375 planning cells across NCT Delhi + fringes	WorldPop 2020 raster scaled to official NCP 2025 projection (22,300,000).

Methodological Boundaries & Disclosures: Boundary Statement on OpenStreetMap Provenance: Crowdsourced OSM POIs and road segments are strictly restricted to the Delhi pilot geography where completeness was empirically validated (3,534 POIs, 8,087 road segments). At the national level, GridSight EV does not rely on OSM for demand modeling; national state rankings derive 100% from official MoRTH VAHAN registrations, MoSPI GSDP tables, and NITI Aayog benchmarks.

- **12-Month Temporal Asynchrony:** The primary national supply registry (BEE, Oct 2025) predates demand data (VAHAN, Oct 2026) by 12 months, naturally exaggerating apparent deficits. GridSight EV mitigates this by providing live Switch Delhi (Oct 2026) as an alternate baseline and surfacing temporal disclaimers.
- **Substation Proximity ≠ Electrical Hosting Capacity:** 99.3% of OSM substations lack MVA capacity tags. In our model, distance to substation serves strictly as a physical interconnection proximity proxy, never claiming unverified electrical capacity.
- **Active Non-Compliant Registrations (26.4%):** Of 10.69M active EVs, 2.82M lack active fitness/PUC. The platform displays both compliant-only (7.87M) and total active fleet metrics, establishing lower and upper demand boundaries.
- **Absence of Real-Time Utilization Telemetry:** Public registries omit real-time connector occupancy and operational uptime. The platform models physical infrastructure presence; OMC operational availability ratios (82.0%) can be applied as a sensitivity discount.

6. Phased Implementation Roadmap, References & Rubric Alignment

Phase	Timeline	Core Objectives & Technical Deliverables	Target Exit Benchmark
Phase 0: Baseline Freeze & AI Integration	Completed (Round 1)	Data pipeline, 28/28 test assertions, Random Forest demand ML, Bass Diffusion forecasting, constrained siting optimizer, zero-watermark UI.	Operational prototype deployed on main branch; 0 TypeScript errors.
Phase 1: Highway Corridor Scaling	Weeks 1–2 (Post-R1)	Expand 5km spatial buffer along NH-48 (Delhi–Jaipur) and Yamuna Expressway using PM Gati Shakti & NHAI freight vector layers for inter-city infill.	Inter-city highway corridor infill & fast-charging simulation tool.
Phase 2: Distribution Grid Capacity Proxies	Weeks 3–4 (Post-R1)	Incorporate secondary CEA distribution standards and DISCOM voltage levels (11kV, 33kV, 66kV) to model feeder hosting headroom without proprietary SCADA.	Grid Feasibility Index per 2km cell combining distance & voltage.
Phase 3: Final Deployment & Grand Finale	Weeks 5–6 (Post-R1)	Containerize full application via Docker; establish high-availability cloud hosting with full offline caching fallback for jury defense.	Rehearsed live jury demonstration with zero external API dependencies.

Official Techfest Evaluation Criteria Alignment Matrix

Evaluation Criterion	Weight	Methodology Document Sections	Repository Verification Evidence
Problem Understanding & Relevance	25%	Section 1: Multi-registry overlap, Burari RTO distortion, spatial vs. volumetric adequacy disconnect.	DATA_AVAILABILITY_AUDIT.md, DATA_GAPS.md, non-additivity design matrix in UI.
Technical Approach & Feasibility	25%	Sections 2 & 3: 375-cell grid, Random Forest ML, continuous Sigmoid decay, Bass diffusion, 2030 energy balance.	scripts/train_ai_models.py, src/utils/planningEngine.ts, 28/28 assertions pass.
Innovation & Differentiation	20%	Section 3 & 4: Physics-Informed ML, greedy submodular siting optimizer, power substitution matrix, dual VAHAN tags.	src/components/AiStudioView.tsx, public/data/ai_optimization_results.json .
Implementation Roadmap & Feasibility	20%	Section 1 & 6: Phase 0 completed prototype, Phases 1–3 highway and DISCOM milestones, exit benchmarks.	DELIVERABLE_AUDIT.md, production build in 7.68s, zero type errors.
Data Strategy & Provenance	10%	Section 5 & References: 8-source provenance table, 19 Oct 2026 freeze, OSM boundary statement, caveats.	data/source_manifest.csv, DATA_QUALITY_REPORT.md, raw data archives.

Key Regulatory & Scientific References

[1] BEE, MoP: Public Charging Stations Data till 26 Oct 2025 (EV_PCS_Data_29277.pdf), Pub. 17 Feb 2026. [2] MoPNG / MHI: PIB PRID 2246042 & 2246045 (OMC EVPCS & FAME-II Deployments), 27 Mar 2026. [3] GNCTD Transport Dept & DDC: Switch Delhi Charging & Swapping Registry (ev.delhi.gov.in), snapshot 19 Oct 2026. [4] MoRTH / NIC: VAHAN 4 Public Analytics Portal (analytics.parivahan.gov.in), snapshot 19 Oct 2026. [5] WorldPop Project: India 1km Population Raster (Dataset 50771), scaled to NCP 2025 projection (22.3M). [6] OpenStreetMap Contributors: Overpass API extracts for Delhi & India stations, highways, substations (ODbL 1.0), snapshot 19 Oct 2026. [7] NITI Aayog & WRI India: India Electric Mobility Index (IEMI) 2025, Pub. Dec 2025. [8] DataMeet Spatial Community: Census-2011 State & District Boundaries (CC BY 4.0). [9] Bass, F. M.: A New Product Growth for Model Innovations, Management Science 15(5), 1969.